Roll No. ....

TEE–611

B. TECH. (EE) (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

SPECIAL ELECTRICAL MACHINE

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Discuss the working of stepper motor with suitable diagram. (CO3)

OR

(b) Enumerate the advantages and disadvantages of VRSM. (CO3)

2. (a) Differentiate between the single phase and two phase on mode operation of stepper motor. (CO2)

OR

(b) State the design parameters for stepper motor. (CO2)

3. (a) Draw the speed torque characteristics of stepper motor and the applications of stepper motor. (CO1)

OR

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(b) Define the following terms : (CO1)

Slew range, Pull in/out torque and Pull in/out rate, Slew range

4. (a) Draw the *two* types of static characteristics of stepper motor and explain.

(CO1)

OR

(b) Write the merits and demerits of permanent magnet stepper motor.

(CO1)

5. (a) Draw the controller block diagram of Stepper motor. (CO4)

OR

(b) Explain the working of (8/6) switched reluctance motor. (CO4)

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TOE–603

B. TECH. (EE) (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

INDUSTRIAL AUTOMATION

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Define industrial automation and explain its significance in modem manufacturing processes. Discuss at least three advantages of implementing automation in industrial settings. (CO1, CO2)

OR

(b) Describe the components of a typical process control system. How do sensors, actuators, controllers, and final control elements collaborate to regulate industrial processes efficiently. (CO1, CO2)

2. (a) What role does a Programmable Logic Controller (PLC) play in industrial automation ? Provide examples of industries processes where PLCs are commonly used, and explain their advantages over traditional relay-based control systems. (CO1, CO2)

P. T. O.

OR

(b) Define SCADA systems and explain their functions in industrial automation. (CO1, CO2)

3. (a) Discuss the significance of distributed control systems (DCS) in modern industrial automation. What are the key features and benefits of DCS, and how do they enhance process control and optimization in large-scale manufacturing plants. (CO1, CO2)

OR

(b) Explain the role of human-machine interface (HMI) in industrial automation systems. How does graphical interfaces provided by HMIs enhance operator interaction, monitoring, and control of industrial processes ? (CO1, CO2)

4. (a) Define data communication components within a PLC system and discuss their roles in facilitating seamless communication between PLCs and other devices in an industrial network. (CO1, CO2)

OR

(b) Explain the scalability and flexibility of SCADA systems in adapting to changing industrial requirements and expanding infrastructure. How can SCAD systems be customized and integrated with other technologies to meet the specific needs of different industries. (CO1, CO2)

5. (a) Discuss the challenges associated with PLC programming and troubleshooting. What strategies can be employed to streamline PLC programming, debugging, and maintenance processes in industrial automation projects ? (CO1, CO2)

OR

(b) Compare and contrast different programming languages used in PLCs, such as ladder logic, function block diagrams, and structured text. What are the specific advantages and disadvantages of each programming language in industrial automation ? (CO1, CO2)

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TEE–602

B. TECH. (EE) (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

POWER SYSTEM—II

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Show that total reactance of the transformer (in per unit system) referred to primary and referred to secondary are same. (CO1)

OR

(b) A 10 kVA 800/400 V, 50 Hz single-phase transformer has primary and secondary leakage reactance each of $5\,\Omega$. Determine total reactance in p.u. (CO1)

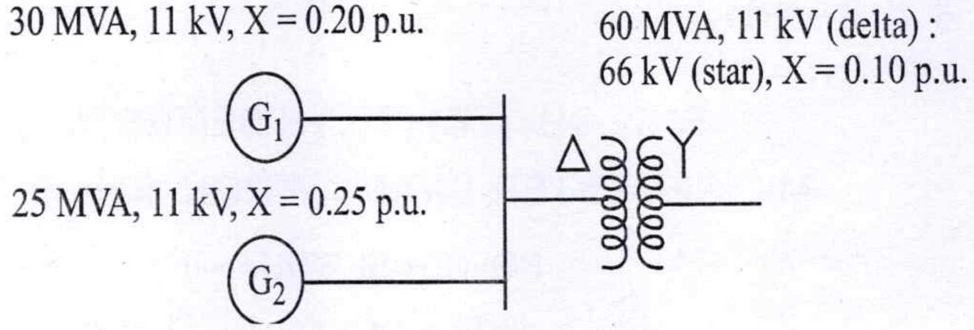
2. (a) Discuss symmetrical component analysis of three-phase unbalanced system. Also derive symmetrical components in terms of unbalanced voltage phasors. (CO1)

OR

(b) Assuming S bases of 25 and 60 MVA, calculate the through impedance in ohms between the generators and the output terminals of the

P. T. O.

transformer for the system, as shown in figure, given below. Specifications of the generators and transformer are indicated in figure : (CO2)



3. (a) Derive the complex power in terms of symmetrical components. (CO2)

OR

(b) In a three-phase four wire system the currents in the lines *a, b* and *c* under abnormal conditions of loading were as follows : (CO2)

$$I_a = 50\angle 45°, \; I_b = 40\angle 300° \text{ and } I_c = 30\angle 180°$$

Calculate the zero, positive and negative phase sequence currents in line a and the return current in the neutral conductor.

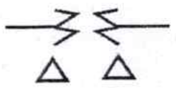
4. (a) Apply symmetrical component analysis and interconnect the sequence networks in case of LLG faults. (CO3)

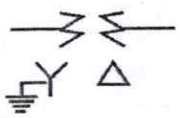
OR

(b) A three-phase synchronous generator delivers 10 MVA at a voltage of 10.5 kV. The line impedance is $5\,\Omega$. Determine the voltage drop in the line in p.u. and in volts. Use the reference base as 12 MVA at 11 kV.

(CO3)

5. (a) Draw zero sequence network for the following transformers : (CO3)

(i) 

(ii) 

(iii) 

OR

(b) A 50 MVA, 11 kV three-phase synchronous generator was subjected to different types of faults. The fault currents are as follows : (CO3)

LG Fault–4200 A; LL Faults–2600 A; LLL Faults–2000 A

The generator neutral is solidly grounded. Find the per unit values of three sequence reactance of the generator.

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TEE–601

B. TECH. (EE) (SIXTH SEMESTER)
MID SEMESTER EXAMINATION, March, 2024

POWER ELECTRONICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Define latching current and holding current as applicable to an SCR. Show these currents on its static I-V characteristics. (CO1, CO2)

OR

(b) With the help of waveform explain the Turn off characteristics of SCR. (CO1, CO2)

2. (a) Explain the protection methods of SCR. (CO1, CO2)

OR

(b) Explain the gate input characteristics of a batch of SCR indicating the upper and lower limit loci and explain why this variation exists. (CO1, CO2)

3. (a) A SCR operating from a peak supply voltage of 400 V has the following specification : (CO1, CO2)

Repetitive peak current is 200 A, (*di*/*dt*) specific is 50 A/us, *dv*/*dt* specific is 200 v/us, choosing the factor of safety is 2. Design the suitable snubber circuit. The minimum value of load resistance is 10 ohm. (CO1, CO2)

P. T. O.

OR

(b) What do you understand by snubber circuit and its requirement during turn on and turn off the SCR operation during series operation.

(CO1, CO2)

4. (a) Explain the single phase half wave control rectifier for RL Load with suitable waveform and circuit diagram. Calculate its Average output voltage. (CO1, CO2)

OR

(b) A dc battery is charged through a resistor R using single phase half wave rectifier configuration having source voltage 230 V, 50 Hz. Calculate the average charging current for R is 8 ohm and E is 150 V. Also find the power supplied to battery and that dissipated in the resistor. (CO1, CO2)

5. (a) A SCR has half cycle surge current rating of 3000 A for 50 Hz supply. Calculate its one cycle surge current rating. (CO1, CO2)

OR

(b) Explain the single phase full wave controlled rectifier with RL load with firing angle 60 degree. Also calculate the average output voltage.

(CO1, CO2)

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TEE–603

B. TECH. (EE) (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

POWER SYSTEM PROTECTION

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Why overlapping of protective zones is desirable ? (CO1)

OR

(b) Describe the desirable features of a power system protection system. (CO1)

2. (a) Why instrument transformers are required in protective relaying scheme ? (CO1)

OR

(b) Derive the expression for maximum momentary current in case of sudden short circuit of transmission line. (CO1)

3. (a) Define the terms Plus Setting Multiplier and time multiplier setting as used in the context of an overcurrent relay. (CO2)

P. T. O.

OR

(b) How are overcurrent relays classified, depending upon the time of operation ? (CO2)

4. (a) Discuss the operating principle of electromagnetic induction relay. (CO2)

OR

(b) An overcurrent relay of 5A is used to protect a feeder through 1000/5 A CT the relay has a PS of 50% and TMS = 0.8. Find the operation of the said relay if a fault current of 1200 A flows through the feeder. Make use of the following characteristic : (CO2)

PSM	**Time for unity TMS (sec)**
1.3	30
2	10
4	6.5
6	3.5
10	3
20	2.2

5. (a) A transmission of inductance 0.2 H and resistance 10 ohm is suddenly short-circuited at the far end. Write the expression for short circuit current $i(t)$. Consider $v = 100 \sin (100\, \pi t + 20)$. (CO3)

OR

(b) Describe the construction and working of a Buchholz relay. (CO3)

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TEE–608

B. TECH. (EE) (SIXTH SEMESTER)

MID SEMESTER EXAMINATION, March, 2024

HIGH VOLTAGE ENGINEERING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Discuss the process of ionization. What is corona discharge ? (CO1)

OR

(b) Explain the Townsend criteria of the breakdown of gaseous dielectric. (CO1)

2. (a) Explain the voltage multiplier circuit and discuss its significance. (CO2)

OR

(b) List the various High voltage set up used for testing, and explain any *one* in detail. (CO2)

3. (a) What are the different processes involved in the study of the breakdown of solid dielectrics ? (CO1)

P. T. O.

OR

(b) Discuss the role of Gases as an insulating media in detail. What is impulse voltage ? (CO1)

4. (a) Explain the Suspended Particle Theory for liquid dielectrics in detail. (CO1, CO2)

OR

(b) Explain the working principle of the electrostatic voltmeter in detail. (CO1, CO2)

5. (a) In an experiment in a certain gas, it was found that the steady-state current is 3.2×10^{-8} A at 8 kV at a distance of 0.5 cm between the plane electrodes. Keeping the field constant and reducing the distance to 0.2cm results in a current of 3.2×10^{-9} A. Calculate Townsend's primary ionization coefficient α. (CO1, CO2)

OR

(b) Explain the Van de Graff Generator. (CO1, CO2)